SER 516 Assignment 4

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Ans1: Here the author refers to fluidity as interchangeability. How easy it is to change the underlying infrastructure /technology and just swap parts. Cloud native applications are meant to be highly configurable and modular. We can see those certain aspects in current software like: Amazon.com/AWS getting more storage or more throughput on days when it is expected to have more traffic e.g. black Friday sales. They can scale the architecture up or down depending on the current requirement. Now with that you also want to have as much redundancy as possible, if some thing fails, there exists a backup or a backup of a backup that can help or replace the downed service instantly. You want such tenacity and redundancy from your cloud native apps to ensure it does not go down at any time.

Ans2: Tying in to the above Ans, anything and everything is more failure prone due to the massive volume each component has to handle. Lets take the example of the Amazon black Friday sales. According to Chain Store Age[1] Amazon has a 40% increase in traffic compared to none sale days and when we see that Amazon had around 120 million site visits[2] on just Black Friday we see that each and every component is under massive stress and is in a position that it cannot fail. Normal PC hardware like storage drives, internet connections and motherboards can simply fail at times but when it is under that much usage and that much stress, failure becomes a certainty due to heat or any other reason. All we can do is ensure that that failure does not affect the customer at any point of time. Hence redundancy becomes very important.

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[1] Berthiaume, Dan. "EXCLUSIVE: Amazon Drives Successful Black Friday Week for Retail." *Chain Store Age*, 6 Dec. 2024, chainstoreage.com/exclusive-amazon-drives-successful-black-friday-week-retail. Accessed 17 Apr. 2025.

[2] Risley, James. "Walmart and Others Lose Cyber 5 Traffic Share." *Digital Commerce 360*, 5 Dec. 2024, www.digitalcommerce360.com/2024/12/05/walmart-loses-traffic-share-over-cyber-5/. Accessed 17 Apr. 2025.

Based on Figure 1:

Ans1:

Microservices: M's, A's and D's are micro services as they are very simple, single task focused services using API's.

Appliances: Databases services and the AWS queue are appliances in this example.

Ans2: There are 3 metrics to observability :

1)Logs: Detailed logs must be maintained to ensure all of the necessary information is present when something fails or goes wrong. To that extent, keeping a standard for logs becomes important as each service and each infrastructure will need its own logging system and they might have to store information that is not needed or available on other services. These might also be spread across the different systems in different location. Saving these logs is also a concern as to what date is a log useful or valid? This needs to be answered as some services may need logs for a longer period than others.

2)Metrics: These values are often used to visually or automatically check the health of the system. The cause for concerns could be from how detailed these values are to what services need to be covered. Details like in the event of a new database turning on or connecting to the service, you want to ensure that you can see the metrics for that database correctly and promptly so that there is no information lost while it is being used. This can spinoff to how do you connect a newly started server immediately to the dashboard.

3)Traces: Traces tell us which component took how much time in the event of a slowdown. This is will always be useful and as such bring some complication with it, such as tracing a request through multiple different services with different setup or trace requirements. Questions can arise such as do we need to trace ALL of the traffic that we receive or do we select some requests for tracing.